# Sigma-Algebras

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**Definition 0.1** ($\sigma$-algebra ($\sigma$-field, Borel field))**.** A **$\sigma$-algebra** [1] (also called a **$\sigma$-field** [2] or a **Borel field** [3], Definition 1.2.1) on a set $S$ is a collection $\mathcal{S}$ of subsets of $S$ that satisfies:

- $\mathcal{S}$ contains $S$ itself.
- For each set $A$ in $\mathcal{S}$, $\mathcal{S}$ contains its complement $S \setminus A$.
- For each sequence $A_1, A_2, \ldots$ of sets in $\mathcal{S}$, $\mathcal{S}$ contains their union $\bigcup_{i=1}^{\infty} A_i$.

Remark ("Borel" in "Borel field")

*Remark* 0.1 ("Borel" in "Borel field"). The "Borel" in "Borel field" is a different use of "Borel" from the Borel $\sigma$-algebra on $\mathbb{R}$, which is one particular $\sigma$-algebra.

**Example 0.1** ($\sigma$-algebras for die rolls)**.** For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ (sets of die rolls), the collection of all subsets of $D$ is a $\sigma$-algebra on $D$, since complements and unions of subsets of $D$ are again subsets of $D$. So is the smaller collection

$$\{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\}$$

which contains $D$, the complement of each of its sets, and every union of its sets: for example, $\{2, 4, 6\} \cup \{1, 3, 5\} = D$.

**Example 0.2** (A collection that is not a $\sigma$-algebra)**.** The collection $\{\emptyset, \{1\}, D\}$ is not a $\sigma$-algebra on $D = \{1, 2, 3, 4, 5, 6\}$: it contains $\{1\}$ but not its complement $D \setminus \{1\} = \{2, 3, 4, 5, 6\}$.

**Example 0.3** (A $\sigma$-algebra is closed under complements and countable unions)**.** A $\sigma$-algebra $\mathcal{S}$ on a set $S$ is closed under complements in $S$ and under unions of sequences of

its sets, because Definition 0.1 requires both. The collection in Example 0.2 is not closed under complements.

**Theorem 0.1** (Closure properties of a $\sigma$-algebra)**.** *If $\mathcal{S}$ is a $\sigma$-algebra on a set $S$, then $\mathcal{S}$ contains:*

- *the empty set $\emptyset$;*
- *the union $A_1 \cup \cdots \cup A_n$ of any finitely many sets $A_1, \ldots, A_n$ in $\mathcal{S}$;*
- *the intersection $\bigcap_{i=1}^{\infty} A_i$ of any sequence $A_1, A_2, \ldots$ of sets in $\mathcal{S}$.*

Proof

*Proof. Empty set.* $\mathcal{S}$ contains $S$, so it contains the complement of $S$, which is $S \setminus S = \emptyset$.
*Finite unions.* Extend $A_1, \ldots, A_n$ to a sequence by setting

$$\begin{aligned} A_{n+1} &= A_{n+2} \\ &= \cdots \\ &= \emptyset; \end{aligned}$$

each of these sets is in $\mathcal{S}$, by the first part. Then:

$$\begin{aligned} A_1 \cup \cdots \cup A_n &= A_1 \cup \cdots \cup A_n \cup \emptyset \cup \emptyset \cup \cdots && \text{(a union with } \emptyset \text{ adds no elements)} \\ &= \bigcup_{i=1}^{\infty} A_i && (A_i = \emptyset \text{ for } i > n) \end{aligned}$$

and $\mathcal{S}$ contains $\bigcup_{i=1}^{\infty} A_i$ by the union rule of Definition 0.1.
*Countable intersections.* Each $A_i$ is a subset of $S$, so an element of $S$ is outside $S \setminus A_i$ exactly when it is in $A_i$; that is, $S \setminus (S \setminus A_i) = A_i$. So, by De Morgan's laws:

$$\begin{aligned} S \setminus \bigcup_{i=1}^{\infty} (S \setminus A_i) &= \bigcap_{i=1}^{\infty} (S \setminus (S \setminus A_i)) && \text{(De Morgan's laws)} \\ &= \bigcap_{i=1}^{\infty} A_i && (S \setminus (S \setminus A_i) = A_i) \end{aligned}$$

Each $S \setminus A_i$ is in $\mathcal{S}$ by the complement rule, so their union is in $\mathcal{S}$ by the union rule, and the complement of that union is in $\mathcal{S}$ by the complement rule again.
So $\mathcal{S}$ is closed under finite unions and under intersections of sequences of its sets. □

**Theorem 0.2** (All subsets form a $\sigma$-algebra)**.** *For any set $S$, the collection of all subsets of $S$ is a $\sigma$-algebra on $S$.*

Proof

*Proof.* Each condition of Definition 0.1 holds:

- $S$ is a subset of itself.
- For each subset $A$ of $S$, the complement $S \setminus A$ contains only elements of $S$, so it is a subset of $S$.
- For each sequence $A_1, A_2, \ldots$ of subsets of $S$, every element of $\bigcup_{i=1}^{\infty} A_i$ is in some $A_i$ and so in $S$; the union is a subset of $S$.

□

**Theorem 0.3** (An intersection of $\sigma$-algebras is a $\sigma$-algebra)**.** *Let $\mathcal{S}_j$ be a $\sigma$-algebra on a set $S$ for each $j$ in a nonempty set of indices $J$. Then their intersection $\bigcap_{j \in J} \mathcal{S}_j$, the collection of subsets of $S$ that are in every $\mathcal{S}_j$, is a $\sigma$-algebra on $S$.*

Proof

*Proof.* Each condition of Definition 0.1 holds:

- $S$ is in every $\mathcal{S}_j$, so it is in the intersection.
- If $A$ is in the intersection, then $A$ is in every $\mathcal{S}_j$, so its complement $S \setminus A$ is in every $\mathcal{S}_j$, and so it is in the intersection.
- If $A_1, A_2, \ldots$ are in the intersection, then they are in every $\mathcal{S}_j$, so their union $\bigcup_{i=1}^{\infty} A_i$ is in every $\mathcal{S}_j$, and so it is in the intersection.

□

**Definition 0.2** ($\sigma$-algebra generated by a collection)**.** Let $\mathcal{C}$ be a collection of subsets of a set $S$. The **$\sigma$-algebra generated by** $\mathcal{C}$, written $\sigma(\mathcal{C})$, is the intersection of all the $\sigma$-algebras on $S$ that contain every set in $\mathcal{C}$.

Remark (The smallest $\sigma$-algebra containing $\mathcal{C}$)

*Remark* 0.2 (The smallest $\sigma$-algebra containing $\mathcal{C}$). $\sigma(\mathcal{C})$ is a $\sigma$-algebra by Theorem 0.3 (the collection of all subsets of $S$ is one of the $\sigma$-algebras intersected, by Theorem 0.2, so there is at least one). It contains every set in $\mathcal{C}$, and it is contained in every other $\sigma$-algebra on $S$ that does, so it is the smallest $\sigma$-algebra on $S$ that contains $\mathcal{C}$.

**Example 0.4** (The $\sigma$-algebra generated by the even rolls)**.** For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$, let $\mathcal{C} = \{\{2, 4, 6\}\}$. Every $\sigma$-algebra on $D$ that contains $\{2, 4, 6\}$ must also contain $D$, the complement $\{1, 3, 5\}$, and $\emptyset$ (Definition 0.1 and Theorem 0.1). The collection $\{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\}$ is already a $\sigma$-algebra (Example 0.1), so it is $\sigma(\mathcal{C})$.

**Definition 0.3** (Borel $\sigma$-algebra)**.** The **Borel $\sigma$-algebra** on $\mathbb{R}$, written $\mathcal{B}$, is the $\sigma$-algebra generated by (Definition 0.2) the collection of all intervals.

**Example 0.5** (The Borel $\sigma$-algebra contains $\mathbb{R}$ and the half-lines)**.** The interval $(-\infty, 0)$ is in $\mathcal{B}$ by Definition 0.3. So is its complement $[0, \infty)$, because $\mathcal{B}$ is a $\sigma$-algebra (Definition 0.1). The set $\mathbb{R}$ is in $\mathcal{B}$ too: a $\sigma$-algebra on $\mathbb{R}$ contains $\mathbb{R}$ itself.

**Definition 0.4** (Borel set)**.** A **Borel set** [4] is a set in the Borel $\sigma$-algebra $\mathcal{B}$.

**Example 0.6** (Some Borel sets)**.**

- Every interval, such as $[0, 1]$, $(2, 5)$, or $[0, \infty)$, is a Borel set.
- A single point $\{a\} = [a, a]$ is an interval, so it is a Borel set.
- The integers are a Borel set: they are the union of the sequence of single points $\{0\}, \{1\}, \{-1\}, \{2\}, \{-2\}, \ldots$ (countable and uncountable sets lists them this way), and $\mathcal{B}$ contains that union by Definition 0.1.
- The complement $\mathbb{R} \setminus [0, 1]$ is a Borel set, because $\mathcal{B}$ contains the complement of each of its sets.


[1] Wikipedia contributors, "Sigma-algebra — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/%CF%83-algebra

[2] P. Billingsley, *Probability and measure*, 3rd ed. in Wiley series in probability and mathematical statistics. New York: Wiley, 1995.

[3] G. Casella and R. Berger, *Statistical inference*, 2nd ed. Cengage Learning, 2002. Available: https://www.cengage.com/c/statistical-inference-2e-casella-berger/9780534243128/

[4] Wikipedia contributors, "Borel set — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Borel_set